



Dr. Vicky Mody

Assistant Professor of Medicinal Chemistry,
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Personal Data:

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Education

University of Houston- *Houston, TX*

Ph.D - Organic chemistry,

Research Advisor – Dr. Roman S. Czernuszewicz

April 2003 - May 2007

Thesis Title:

Resonance Raman Scattering by High Valent Oxo Metalloporphyrins

University of Mumbai, *Mumbai, India*

Master of Science - Analytical Chemistry,

Aug 1999 - July 2001

University of Mumbai, *Mumbai, India*

Bachelor of Science - Chemistry

Optional – Drugs and Dyes

Aug 1996 - July 1999

Professional Experience

Post-Doctorate Fellow, Dept. of Pharmacological and Pharmaceutical Sciences, University of Houston,
Houston, TX *May 2007 – July 2009*

Summer Mentorship program University of Houston, Houston, *May 2008 – present.*

Mentored 3 undergraduate students from various discipline for summer 2008. Designed a project for them and trained them in synthesis and characterization of bioresponsive nanoparticles for the early detection of breast and ovarian cancer.

Associate Chemist, Polyorganix Inc. Houston, TX,

May 2005 - July 2007

Teaching Assistant, University of Houston, *Houston, TX* Jan 2003 – Jan 2005.
Taught and supervised Organic and Analytical chemistry laboratory classes. Proctored exams, graded laboratory reports and exams.

Analytical Development chemist, Accutest Research Limited, *Mumbai, India* ,
May 2001 - July 2002

Chemist, Bharat Rasayan Chemicals, *Tarapur, India*. May 2000- Aug 2000

Consulting Experience

Polyorganix Inc. Houston, TX, Design of simpler methods to synthesize of novel biopolymers and light emitting polymers of for commercial use. May 2008 – Present

Peer Review Committee:

- Journal of Controlled Release.
- Journal of Raman Spectroscopy.
- Organic Geochemistry.
- Critical Review in Therapeutic Drug Systems.

Significant Accomplishments:

- Designed and developed novel doped ultrasmall superparamagnetic iron oxide nanoparticles based MRI contrast agents which can be used for the early diagnosis of ovarian, and breast cancers.
- Successfully executed the synthesis of bioresponsive agent for targeting breast and ovarian cancer.
- Efficiency and the sensitivity of these bioresponsive agents are being evaluated using *in vitro* and *in vivo* studies.
- Investigating cytotoxicity of synthesized nanoparticles using cell viability assay (MTT assay, LDH assay) on various cell lines such as MCF-7, 4T1s, PC-3, A7R5, and HDFs.
- Synthesized various monomers and light emitting polymers, biopolymers, using single step and multistep reactions.
- Designed and synthesized poly(phosphazene) based drug delivery system for the controlled release of Ibuprofen.
- Developed and improved, regioselective synthesis of thiazoles featuring a reduced number of synthetic steps compared to current multistep reactions.
- Synthesized various isotopically substituted analogs of imidazole in order to study the effects of hydrogen bonding from N(1)H, and N(2)H protons of histidine on the heme intermediates.
- Synthesized various metalloporphyrins and isotopically substituted metalloporphyrins such as Ni(II), V(IV)O, and Mn(V)N complexes of Proto, meso and deuteroporphyrin.
- Studied the interaction of imidazole and its derivatives with biological porphyrins using resonance Raman spectroscopy.

Publications and Selected Presentations:

- Roman S Czernuszewicz, **Vicky V Mody**, Arkadiusz Czader, Michael G, Volodymyr S, Karol Grela and Daniel T. Gryko, "Why the Chromyl Bond Is Stronger than the Perchromyl Bond in High-Valent Oxochromium(IV,V) Complexes of Tris(pentafluorophenyl)corrole". *JACS*, 2009, 131 (40), 14214-14215.
- **Vicky V. Mody**, Marzena B. Fitzpatrick, Samantha S. Zabaneh, Roman S.Czernuszewicz, Michał Gałezowski, and Daniel T. Gryko, "Solvent effects on the electronic and vibrational properties of high-valent oxomolybdenum(V) 5,10,15-triphenylcorrole probed by UV-visible and resonance Raman spectroscopy" *Journal of Porphyrins and Pthalocyanins*, 2009, 13 1040-1052.
- **Vicky V. Mody**, Mohamed Ismail Nounou, and Bikram Malavosklsh; "Novel Nanomedicine-Based MRI Imaging Systems for Gynecological Malignancies."; Review Article, *Advanced Drug Delivery Reviews*, 2009, 61(10); 795-807.
- Roman S Czernuszewicz, **Vicky V Mody**, Adelajda Zareba, Michael G, Volodymyr S, Karol Grela and Daniel T. Gryko, "Solvent Dependent Resonance Raman Spectra of High Valent oxomolybdenum(V) Tris[3,5,bis-(trifluoromethyl)phenyl] corrolate". *Inorganic Chemistry*, 2007, 46(14); 5616-5624.
- A. Worstynowicz, S. M. Kazmarek, M. Bosacka, **Vicky V Mody**, and R. S. Czernuszewicz. "Structural and Magnetic Characterization of the C^{3+} and Ni^{2+} ion Species in $Ni_2CrV_3O_{11}$ ". *Rev. Adv. Mat. Sci.* 2007, 14, 24-32.
- A. Worstynowicz, S. M. Kazmarek, M. Bosacka, **Vicky V Mody**, and R. S. Czernuszewicz. "Vanadochromates with Divalent Metals; Structural and Magnetic Characterization". *Rev. Adv. Mat. Sci.* 2007, 14, 33-40.
- M. M. Ismail Nounou, **V. Mody**, K. Emmanouil, T. Pham, Z. Lu, and M.Bikram, Design, Synthesis and Characterization of Novel Bio-reducible Nonviral Gene Delivery Systems, Poster Presentation, BMES 2009 Annual Fall Scientific Meeting, Pittsburgh, PA.
- **V. V. Mody**, M. I. Nounou, L. Cui, C. Karmonik, B. Lorenz, and M. Bikram, "Development of a MMP-Specific MRI Contrast Agent for the Early Detection of Ovarian Cancer", Poster Presentation, BMES 2009 Annual Fall Scientific Meeting, Pittsburgh, PA.
- D. E. Gheorghe, **V. V. Mody**, J. M. Penaloza, B. Lorenz, C. Karmonik, and M. Bikram, "Synthesis and Characterization of Gold Alloy Nanoshells on Gd-Doped Iron Oxide Nanoparticles" Poster Presentation, BMES 2009 Annual Fall Scientific Meeting, Pittsburgh, PA.
- **Vicky V. Mody**, Mohamed Nounou, and Bikram Malavosklsh, Synthesis of gold coated doped iron oxide based nanoparticles as the MR contrast agents. *Nanoletters*, *Manuscript under preparation*.
- **Vicky V. Mody**, Mohamed Ismail Nounou, Christof Karmonik, and Malavosklsh Bikram, Development of Tumor-Targeted MRI Contrast Agents for the Early Detection of Ovarian Cancer, Poster Presentation 35th Annual Meeting of Controlled Release Society, New York, NY, 2008.
- **Vicky V. Mody**, Kamilia Abdelraouf, Mohamed Nounou, Christof Karmonik and Malavosklsh Bikram. "Synthesis and characterization of bioresponsive ultrasmall paramagnetic ironoxide nanoparticles as MR contrast agents". **Selected for Podium Presentation** at 25th Annual Meeting of HSEMB, Houston, TX, 77004.
- Malavosklsh Bikram, **Vicky V. Mody**, Mohamed Ismail Nounou, and Christof Karmonik; "Tumor-specific MRI Contrast Agents for Ovarian Cancer"; Poster Presentation, Drug Carriers In Medicine & Biology (Gordon Research Conference), Big Sky, MT, August 24-29, 2008.
- **Vicky V. Mody**, Arkadiusz Czader, and Roman Czernuszewicz, "Interaction of Imidazole with the V(IV)O bond of vanadyl octaethylporphyrin characterized by variable excitation wavelength

resonance Raman scattering”, Presented at 62th Southwest Regional Meeting of the American Chemical Society, Houston, TX, 2006.

- Stephen Potter, **Vicky V. Mody**, Ife Bamgbose, Rithu Johnson, Mohamed Ismail Nounou, and Bikram Malavosklis, Novel doped iron oxide MRI contrast agents for the early detection of ovarian cancer. Poster Presentation, AGEF, 2008, Houston, TX, 77030.
- Ife Bamgbose, **Vicky V. Mody**, Stephen Potter, Rithu Johnson, Mohamed Ismail Nounou, and Bikram Malavosklis; "Synthesis and characterization of a novel biodegradable ovarian cancer targeted MRI contrast agents"; Poster Presentation, SURF, Houston, TX, 2008.

Book Chapter

- **Vicky V. Mody**, “Synthesis of Nanoparticles and its Application in Medical Imaging”, in Science and Commercialization of Nanoparticles, Ed. Waqar Ahmed;, Pan Stanford Publishing, *Under Preparation*.

Awards and Affiliations:

- Secured first place in MSc. at B.N.N college of Bhiwandi, University of Mumbai.
- Registered Member of Houston Society for Engineering in Medicine and Biology.
- Registered Member of Controlled Release Society.